

Big Line or Big Clique in Planar Point Sets

OPG-588

L1 - TRACTABLE

PARTIALLY SOLVED

Geometry

ORIGINAL PROBLEM STATEMENT

Let S be a set of points in the plane. Two points v and w in S are visible with respect to S if the line segment between v and w contains no other point in S .

Conjecture For all integers $k, \ell \geq 2$ there is an integer n such that every set of at least n points in the plane contains at least ℓ collinear points or k pairwise visible points.